

Milestone 1: Dataset Collection & Preprocessing

1. Objective

The goal of this milestone is to collect a dataset of scanned document images from multiple scanner devices, organize and label the dataset, analyze the basic image properties, and perform preprocessing to standardize the dataset for further use in machine learning or image analysis tasks.

2. Dataset Collection

2.1 Source and Structure

- Scanned document samples were collected from **5 different scanner models** (brands).
- Each scanner model folder contains two subfolders corresponding to scanning resolutions: **150 dpi** and **300 dpi**.
- Each resolution subfolder contains approximately **100 scanned images**, resulting in a total of 1,000 images (5 models × 2 resolutions × 100 images per resolution).

2.2 Dataset Organization

The dataset follows a hierarchical folder structure:

```
dataset_scanners/  
|  
├─ Scanner_Model_1/  
|   ├── 150/  
|   |   ├── image_001.tif  
|   |   ├── ...  
|   └─ 300/  
|       ├── image_101.tif  
|       ├── ...  
├─ Scanner_Model_2/  
|   ├── 150/  
|   └─ 300/  
...
```

2.3 Labeling

A CSV file named `scanner_labeled_dataset.csv` was created to store metadata for each image, including:

- `scanner_model`: Name of the scanner brand/model
- `subtype`: Scanning resolution (150 or 300 dpi)
- `file_name`: Name of the image file
- `file_path`: Full path to the image
- `resolution_width` and `resolution_height`: Image dimensions
- `format`: Image file format (e.g., TIFF, PNG)
- `color_mode`: Color channel information (e.g., RGB, L)

Methodology:

- Used Python's `os` library to traverse the folder structure.
 - Utilized `PIL.Image` to extract image metadata.
 - Stored all metadata in a Pandas DataFrame and exported it to a CSV file.
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3. Image Analysis

The following basic image properties were analyzed for all collected images:

1. **Resolution** – Width and height of each image in pixels.
2. **File Format** – Type of image file (e.g., TIFF, PNG, JPEG).
3. **Color Mode** – Whether the image is RGB, grayscale, or other formats.

Purpose:

Understanding these properties helps in designing preprocessing steps and ensures uniformity in the dataset for model training.

4. Image Preprocessing

To prepare the dataset for machine learning tasks, the following preprocessing steps were applied:

4.1 Resizing

- All images were resized to a fixed dimension of **1024 × 1024 pixels** using the **LANCZOS** resampling method.

- Ensures consistent input size for models and reduces variability caused by different scanner resolutions.

4.2 Color Conversion

- All images were converted to **grayscale (L mode)**.
- Simplifies image data by reducing the number of channels from 3 (RGB) to 1 while preserving essential texture information.

4.3 Denoising

- Optional denoising can be performed to remove scanner-specific noise.
- Not implemented in this milestone but considered for future preprocessing steps.

4.4 File Handling

- Images were read using **tifffile** to handle TIFF files properly.
- Preprocessed images were saved into a structured output folder that mirrors the input dataset hierarchy:

```
preprocessed_images/  
├─ Scanner_Model_1/  
│   ├─ 150/  
│   └─ 300/  
...
```

5. Dataset Normalization and Structure

- The preprocessed dataset is normalized in terms of resolution and color.
- Folder and CSV-based labeling make it easy to reference each image with its scanner source and resolution.
- This structured format ensures the dataset is ready for further steps such as **feature extraction, model training, and classification**.

6. Summary

- **Total Images Collected:** 1,000 (approx.)

- **Number of Scanner Models:** 5
- **Preprocessing Performed:** Resizing to 1024×1024, grayscale conversion
- **Output:**
 - Preprocessed images saved in `preprocessed_images/`
 - Metadata CSV saved as `scanner_labeled_dataset.csv`